

Figure 1: Colour from tuple (123,212,255)

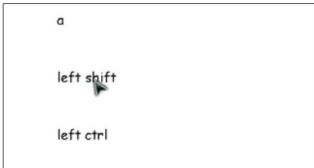


Figure 2: Result of <Left-Ctrl><Left-Shift><a>

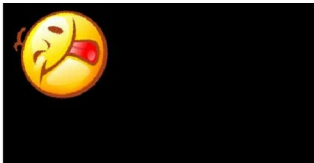


Figure 3: Rotated image



Figure 4: Simple sprite sheet

## Rotation

Rotation is the key to making any shooting game. You can rotate the image both with the keyboard as well as the mouse—but, generally, rotation is done with the mouse, because movements are done using the keyboard. Rotation in Pygame can be done by using its *transform* module. The script below demonstrates rotation with both, keyboard and mouse.

```
#rotation.py
import pygame
from pygame.locals import *
from sys import exit
```

```
pygame.init()
screen=pygame.display.set_mode((640,480),0,0)
pygame.display.set_caption("Rotation using transform
module")
image=pygame.image.load("naughty.png").convert_alpha()
image2=image
a=0
clock=pygame.time.Clock()
#1
while True:
    for i in pygame.event.get():
        if i.type==QUIT:
            exit()
        screen.fill((0,0,0))
        rotation=pygame.mouse.get_rel()
    #2
    buttonpress=pygame.mouse.get_pressed()
    #3
    press=pygame.key.get_pressed()
    screen.blit(image2,(100-(image2.get_width()/2),100-(image2.
get_height()/2)))    #4
    if rotation[0] and buttonpress[0]:
        a=a+rotation[0]
        image2=pygame.transform.rotate(image, a)
    #5
    if press[K_LEFT]:
        a=a-10
        image2=pygame.transform.rotate(image, a)
    if press[K_RIGHT]:
        a=a+10
        image2=pygame.transform.rotate(image, a)
    pygame.display.update()
    clock.tick(15)
```

Run the script and use the Left/Right arrow keys, or left-click and drag, to rotate the naughty-face icon, as shown in Figure 3. Let's understand the code. We loaded the *naughty.png* image into the video display, and then we performed the following steps:

1. The time module's *Clock()* function is used to determine the speed of the game through frames. When used with the *tick(value)* function (see the last line of the code), it decides how many frames per second should appear on the video display. When the value is 1, the frame rate is 1 per second. As you increase the value, the speed or frame-rate of the game increases. Thirty frames per second is recommended as the best frame rate for games, as every hardware is capable of providing this rate continuously.
2. *pygame.mouse.get\_rel()*: It returns a tuple of the change in co-ordinates with respect to the previous position of the mouse.